7. Hypothesis Testing

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Outline

1. Introduction to Hypothesis Testing
2. The Errors of Our Ways
3. Normal Mean Test with Known Variance
4. Normal Mean Test with Known Variance: Design
5. Two-Sample Normal Means Test with Known Variances
6. Normal Mean Test with Unknown Variance
7. Two-Sample Normal Means Tests with Unknown Variances
8. Two-Sample Normal Means Test with Paired Observations
9. Normal Variance Test
10. Two-Sample Normal Variances Test
11. Bernoulli Proportion Test
12. Two-Sample Bernoulli Proportions Test
13. Goodness-of-Fit Tests: Introduction
14. Goodness-of-Fit Tests: Examples
15. Goodness-of-Fit Tests: Honors Example

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This is a **two-sided test**. We will reject the belief of the null hypothesis H_0 if $\hat{\mu}$ (an estimator of μ) is "too high" or "too small."

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This is a **one-sided test**. We'll reject the null hypothesis if $\hat{\mu}$ is "too small," and only then will make the claim that the emissions are low.

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If you get substantial evidence supporting H_1, you'll decide to reject H_0. Otherwise, you "fail to reject" H_0. (This roughly means that you grudgingly accept H_0.)

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Lots more details later.

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   – If the probability from (d) is low, reject H_0 and select H_1.
   – If the probability from (d) is high, fail to reject H_0.

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The test statistic is

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Equivalently, I'd reject H_0 if t_{obs} is "significantly" $\ll 0$.

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If yes, then we we'll *fail to reject* ("grudgingly accept") H_0.

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If no, then we'll *reject* H_0 in favor of H_1.

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This means that

$$0.05 \;<\; p \;\equiv\; P(t(19) < \underbrace{-1.61}_{t_{\text{obs}}}) \;<\; 0.10.$$

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More on this pretty soon! $\square$

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Where Can We Go Wrong? Four things can happen:

- If H_0 is actually true and we conclude that it's true — good. ☺
- If H_0 is actually false and we conclude that it's false — good. ☺
- If H_0 is actually true and we conclude that it's false — bad.

Lesson 7.2 — The Errors of Our Ways

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- If H_0 is actually true and we conclude that it's true — good. ☺
- If H_0 is actually false and we conclude that it's false — good. ☺
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It's good to have high power.

1. Introduction to Hypothesis Testing
2. The Errors of Our Ways
3. Normal Mean Test with Known Variance
4. Normal Mean Test with Known Variance: Design
5. Two-Sample Normal Means Test with Known Variances
6. Normal Mean Test with Unknown Variance
7. Two-Sample Normal Means Tests with Unknown Variances
8. Two-Sample Normal Means Test with Paired Observations
9. Normal Variance Test
10. Two-Sample Normal Variances Test
11. Bernoulli Proportion Test
12. Two-Sample Bernoulli Proportions Test
13. Goodness-of-Fit Tests: Introduction
14. Goodness-of-Fit Tests: Examples
15. Goodness-of-Fit Tests: Honors Example

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Notice that a lower α results in a higher $z_{\alpha/2}$. Then it's "harder" to reject H_0. For instance, if $\alpha = 0.01$, then $z_{0.005} = 2.58$, and we would *fail to reject* H_0 in this example. $\square$

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1. Introduction to Hypothesis Testing
2. The Errors of Our Ways
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4. Normal Mean Test with Known Variance: Design
5. Two-Sample Normal Means Test with Known Variances
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$$n \doteq \frac{\sigma^2}{\delta^2}(z_{\alpha/2} + z_\beta)^2 = \frac{16}{4}(1.96 + 1.645)^2 = 51.98.$$

Recap: If you want to test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$, and

(1) You know σ^2,

(2) You want $P(\text{Type I error}) = \alpha$, and

(3) You want $P(\text{Type II error}) = \beta$ when $\mu = \mu_1 (\neq \mu_0)$,

then you have to take $n \doteq \sigma^2 (z_{\alpha/2} + z_\beta)^2 / \delta^2$ observations.

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$$n \doteq \frac{\sigma^2}{\delta^2}(z_{\alpha/2} + z_\beta)^2 = \frac{16}{4}(1.96 + 1.645)^2 = 51.98.$$

In other words, we need about 52 observations. $\square$

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the p-value is $p = 2\big(1-F_{n-1}(|T_0|)\big)$. $\square$

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Alternatively, note that the p-value is $2\big(1 - F_{14}(2.07)\big) = 0.0574$, where the Excel function `T.DIST` can be used to evaluate the cdf $F_{14}(2.07)$. Since $p = 0.0574 > 0.05 = \alpha$, we didn't quite reject. $\square$

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2. The Errors of Our Ways
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4. Normal Mean Test with Known Variance: Design
5. Two-Sample Normal Means Test with Known Variances
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7. Two-Sample Normal Means Tests with Unknown Variances
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Paired t-test: (X_i, Y_i) observations paired (next lesson).

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$$\bar{X} \equiv \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad \bar{Y} \equiv \frac{1}{m}\sum_{i=1}^{m} Y_i$$

$$S_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \quad \text{and} \quad S_y^2 = \frac{1}{m-1}\sum_{i=1}^{m}(Y_i - \bar{Y})^2.$$

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As in the previous module, define the **pooled variance estimator** by

$$S_p^2 \equiv \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}.$$

If H_0 is true, it can be shown that

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Thus,

$$\text{Reject } H_0 \quad \text{iff} \quad |T_0| > t_{\alpha/2, n+m-2}.$$

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Example: Catalyst X is currently used by a certain chemical process. If catalyst Y gives higher mean yield, we'll use it instead.

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Suppose we have the following data:

$$n \;=\; 8, \;\; \bar{X} \;=\; 91.73, \;\; S_x^2 \;=\; 3.89$$

$$m \;=\; 8, \;\; \bar{Y} \;=\; 93.75, \;\; S_y^2 \;=\; 4.02.$$

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Thus, we should probably use catalyst Y. $\square$

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The following table summarizes how to carry out the various two- and one-sided tests.

two-sided	$H_0 : \mu_x = \mu_y$ $H_1 : \mu_x \neq \mu_y$	Reject H_0 iff $\lvert T_0^\star \rvert > t_{\alpha/2,\nu}$

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one-sided	$H_0 : \mu_x \geq \mu_y$ $H_1 : \mu_x < \mu_y$	Reject H_0 iff $T_0^\star < -t_{\alpha,\nu}$

Example: Let's test $H_0 : \mu_x = \mu_y$ vs. $H_1 : \mu_x \neq \mu_y$ at level $\alpha = 0.10$.

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Since $|T_0^\star| < t_{\alpha/2,\nu}$, we *fail to reject* (i.e., grudgingly accept) H_0.
Actually, since $T_0^\star$ was so close to 0, we didn't really need the tables. $\square$

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Define the sample mean and variance of the differences,

$$\bar{D} \equiv \sum_{i=1}^{n} D_i/n \sim \text{Nor}(\mu_d, \sigma_d^2/n)$$

$$S_d^2 \equiv \frac{1}{n-1} \sum_{i=1}^{n} (D_i - \bar{D})^2 \sim \frac{\sigma_d^2 \chi^2(n-1)}{n-1}.$$

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So we *fail to reject* H_0. $\square$

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Assume all X's and Y's are independent.

We'll estimate the variances σ_x^2 and σ_y^2 by the sample variances S_x^2 and S_y^2.

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$$F_0 < F_{1-\alpha/2,n-1,m-1} \quad \text{or} \quad F_0 > F_{\alpha/2,n-1,m-1}.$$

iff (because of a property of the F distribution discussed earlier)

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Since $F_{1-\alpha/2,n-1,m-1} \leq F_0 \leq F_{\alpha/2,n-1,m-1}$, we *fail to reject* H_0. $\square$

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2. The Errors of Our Ways
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4. Normal Mean Test with Known Variance: Design
5. Two-Sample Normal Means Test with Known Variances
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2. The Errors of Our Ways
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Denote the respective sample means by

$$\bar{X} \;=\; \frac{1}{n}\sum_{i=1}^{n} X_i \;\sim\; \frac{\text{Bin}(n, p_x)}{n} \quad \text{and} \quad \bar{Y} \;\sim\; \frac{\text{Bin}(m, p_y)}{m}.$$

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$H_0 : p_x \ge p_y$ vs. $H_1 : p_x < p_y \quad \Rightarrow \quad$ reject H_0 iff $Z_0 < -z_\alpha$.

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(Why does the above test statistic remind me of Old McDonald's Farm? ☺)

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- Other g-o-f tests: Kolmogorov–Smirnov, Anderson–Darling, Shapiro–Wilk, etc.

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Discrete Example: The number of defects in printed circuit boards is hypothesized to follow a Geometric(p) distribution. We collect a random sample of $n = 70$ printed boards, and we observe the number of defects in each.

# defects	frequency
1	34
2	18
3	2
4	9
5	7
	70

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$$\hat{p} = \frac{1}{\bar{X}} = \frac{70}{1(34) + 2(18) + 3(2) + 4(9) + 5(7)} = 0.4762.$$

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*Combine the entries in the last row (≥ 5) so the probabilities sum to one.

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This means that the number of defects probably isn't geometric. $\square$

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The issue is that the a_i's might depend on unknown parameters.

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Further suppose we determine which interval each of the 100 observations belongs to and tally them up to get the O_i's. . . .

interval $(\hat{a}_{i-1}, \hat{a}_i]$	O_i	$E_i = n/k$
[0, 0.092]	0	10
(0.092, 0.196]	1	10
(0.196, 0.313]	1	10
(0.313, 0.448]	6	10
(0.448, 0.608]	17	10
(0.608, 0.804]	21	10
(0.804, 1.057]	23	10
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So reject H_0. These guys ain't Expo. No way, no how. ☹

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3. Normal Mean Test with Known Variance
4. Normal Mean Test with Known Variance: Design
5. Two-Sample Normal Means Test with Known Variances
6. Normal Mean Test with Unknown Variance
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8. Two-Sample Normal Means Test with Paired Observations
9. Normal Variance Test
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11. Bernoulli Proportion Test
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Lesson 7.15 — Goodness-of-Fit Tests: Honors Example

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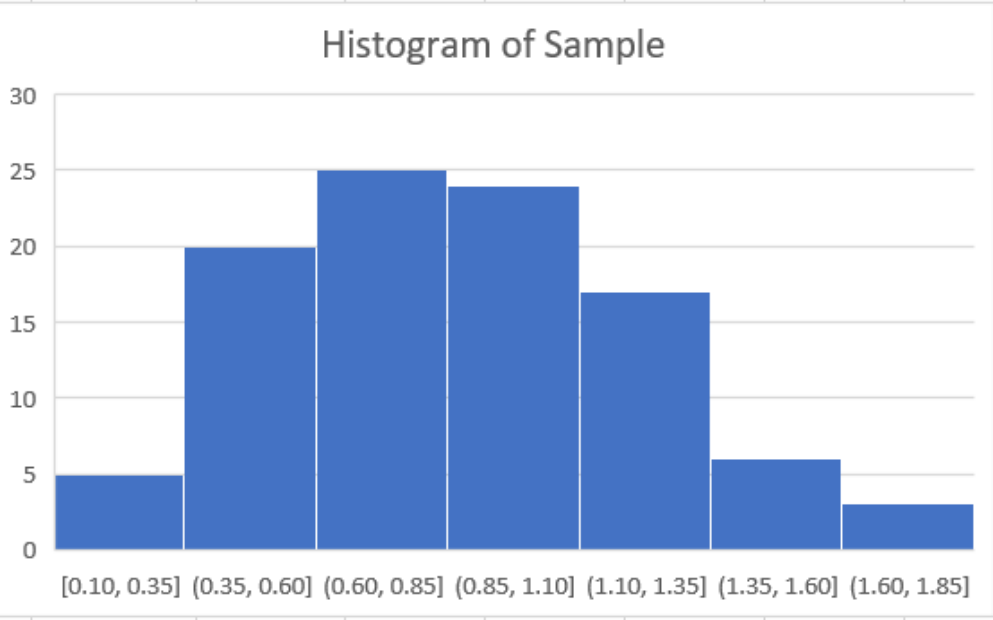

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Along the way, we'll encounter a number of interesting issues that we'll need to deal with; but it'll all work out in the end. ☺

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So this motivates our need to look at other distributions in our quest to find a good data fit.

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and where $\Psi(r) \equiv \Gamma'(r)/\Gamma(r)$ is the *digamma function*.

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- **Newton's method** — stay tuned!

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Each iteration of the algorithm chops the search area in two and therefore converges to $r^\star$ pretty quickly.

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Let's take the approximate differentiation term $h = 0.01$. Then here's what Equation (1) simplifies to:

$$g(r) \quad \doteq \quad 100\,\ell\mathrm{n}(r) - 100\,\ell\mathrm{n}(0.8778) - \frac{100}{0.01}\Big(\frac{\Gamma(r+0.01)}{\Gamma(r)} - 1\Big) - 21.5623$$

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The algorithm is depicted in all of its glory in the next table.

step	ℓ_j	$g(\ell_j)$	u_j	$g(u_j)$	r_j	$g(r_j)$

step	ℓ_j	$g(\ell_j)$	u_j	$g(u_j)$	r_j	$g(r_j)$
0	5.0000	0.5506	7.0000	-3.0595	6.0000	-1.5210

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2	5.0000	0.5506	5.5000	−0.5698	5.2500	−0.0338
3	5.0000	0.5506	5.2500	−0.0338	5.1250	0.2519
4	5.1250	0.2519	5.2500	−0.0338	5.1875	0.1075
5	5.1875	0.1075	5.2500	−0.0338	5.2188	0.0365
6	5.2188	0.0365	5.2500	−0.0338	5.2344	0.0013
7	5.2344	0.0013	5.2500	−0.0338	5.2422	−0.0163
⋮						
14	5.2349	0.0000	5.2349	0.0000	$\boldsymbol{r^\star = 5.2349}$	0.0000

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The (approximate) endpoints of the intervals are implicitly given by $\widehat{F}(\hat{a}_i) = i/k$, $i = 0, 1, 2, \ldots, k$, where $\widehat{F}(x)$ is the cdf of the Gam$(\hat{r}, \hat{\lambda})$ distribution.

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Sadly, gamma distribution's cdf doesn't have a closed-form. But that's why we have Excel (or its friends) around, e.g.,

$$\hat{a}_i \;=\; \widehat{F}^{-1}(i/k) \;=\; \texttt{GAMMAINV}(i/k, \hat{r}, \hat{\lambda}).$$

interval $(\hat{a}_{i-1}, \hat{a}_i]$	O_i	$E_i = n/k$
[0.000, 0.436]	8	10
(0.436, 0.550]	12	10
(0.550, 0.644]	6	10
(0.644, 0.733]	9	10
(0.733, 0.823]	12	10
(0.823, 0.920]	8	10
(0.920, 1.032]	13	10
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So fail to reject H_0. These may indeed be gamma!

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$$\ell\mathrm{n}(L) \;=\; nr\,\ell\mathrm{n}(\lambda) + n\,\ell\mathrm{n}(r) + (r-1)\,\ell\mathrm{n}\Big(\prod_{i=1}^n x_i\Big) - \lambda^r \sum_{i=1}^n x_i^r.$$

At this point, maximize with respect to r and λ by setting

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$$g(r) \;=\; \frac{n}{r} + \ell\mathrm{n}\Big(\prod_{i=1}^{n} x_i\Big) - \frac{n\sum_i x_i^r\,\ell\mathrm{n}(x_i)}{\sum_i x_i^r} \;=\; 0.$$

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To use Newton, we need (after yet more algebra)

$$g'(r) \;=\; -\frac{n}{r^2} - \frac{n\sum_i x_i^r\,[\ell\mathrm{n}(x_i)]^2}{\sum_i x_i^r} + \frac{n\big[\sum_i x_i^r\,\ell\mathrm{n}(x_i)\big]^2}{\big[\sum_i x_i^r\big]^2}.$$

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1	2.8224	0.7748	-23.8654
2	2.8549	0.1170	-23.6493
3	2.8598	0.0178	-23.6174
4	2.8606	0.0027	-23.6126
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Hence, $\hat{r} = r_5 = 2.8607$, and thus,

$$\widehat{\lambda} = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^{\hat{r}}\right)^{-1/\hat{r}} = 1.0148.$$

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So we fail to reject H_0, and we'll grudgingly pretend that these observations are Weibull. $\square$

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(0.6872, 0.7792]	11	10
(0.7792, 0.8669]	8	10
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The Big Reveal: I actually generated the observations from a Weibull distribution with parameters $r = 3$ and $\lambda = 1$. So we did pretty well! ☺